Linear Algebra Foundations for Linear Models

Applied Linear Models I • Lecture 1 of 16

Applied Linear Models I

Readings: Faraway (2015) ch. 2 • Monahan (2008) ch. 1–4
Kutner et al. (2005) ch. 5 • Montgomery et al. (2012) app. C

Where this course is going

- Linear models put the mean response in a plane and recover it by projection.
- Today's algebra is the language for every later topic:
  - WLS & GLS $\rightarrow$ change the inner product;
  - Ridge / Stein $\rightarrow$ shrink toward the origin;
  - GLMs $\rightarrow$ change the link between $X\beta$ and $\mathbb{E}[y]$.
- Companion code (all self-checking): `R/lecture_01/01_worked_examples.R`.

Today's destinations

Model matrix $\rightarrow$ column space $\rightarrow$ normal equations $\rightarrow$ projection (hat matrix) $\rightarrow$ ANOVA = Pythagoras $\rightarrow$ FWL $\rightarrow$ eigenstructure & collinearity.

The linear model in matrix form

For n units and p regressors,

$$y_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i \quad \Longleftrightarrow \quad \underbrace{\mathrm{y}}_{n\times 1} = \underbrace{\mathrm{X}}_{n\times p}\underbrace{\boldsymbol{\beta}}_{p\times 1} + \underbrace{\boldsymbol{\varepsilon}}_{n\times 1}.$$

- "Linear" means linear **in the parameters $\boldsymbol{\beta}$**, not in the regressors.
- $y = \beta_1 + \beta_2 x + \beta_3 x^2 + \varepsilon$ *is* linear; $y = \beta_1 e^{\beta_2 x} + \varepsilon$ is not.
- Polynomials and interactions just add columns to X.

Assumptions: (A1) $\mathbb{E}[\mathrm{y} \mid \mathrm{X}] = \mathrm{X}\boldsymbol{\beta}$; (A2) $\mathrm{rank}(\mathrm{X}) = p \le n$; (A3) $\mathrm{Var}(\boldsymbol{\varepsilon} \mid \mathrm{X}) = \sigma^2 I$; (A4) normal errors (for exact inference). *Lecture 1 uses only (A2).*

The column space $\mathcal{C}(X)$

$$\mathcal{C}(X) = \{X\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\} = \text{span}\{X_{\cdot 1}, \ldots, X_{\cdot p}\}.$$

- y is a point in $\mathbb{R}^n$; the columns of X are vectors in $\mathbb{R}^n$.
- Under (A2), $\mathcal{C}(X)$ is a *p*-dimensional plane through the origin.
- Modelling claim: $\mathbb{E}[y \mid X] \in \mathcal{C}(X)$; noise pushes y off the plane.
- $\dim \mathcal{C}(X) = p, \quad \dim \mathcal{C}(X)^{\perp} = n - p$
  $\Rightarrow$ these are the **degrees of freedom**.

Fitting = nearest point

Find the point in $\mathcal{C}(X)$ closest to y. "Closest" in Euclidean distance gives least squares.

Least squares and the normal equations

Minimize the squared distance to the plane:

$$S(\boldsymbol{\beta}) = \|\mathrm{y} - \mathrm{X}\boldsymbol{\beta}\|^2 = \mathrm{y}^\top \mathrm{y} - 2\boldsymbol{\beta}^\top \mathrm{X}^\top \mathrm{y} + \boldsymbol{\beta}^\top \mathrm{X}^\top \mathrm{X}\boldsymbol{\beta}.$$

Setting the gradient to zero:

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathrm{X}^\top \mathrm{y} + 2\mathrm{X}^\top \mathrm{X}\boldsymbol{\beta} = 0 \quad \Longrightarrow \quad \boxed{\mathrm{X}^\top \mathrm{X}\,\hat{\boldsymbol{\beta}} = \mathrm{X}^\top \mathrm{y}}$$

- Residual is **orthogonal** to every column: $\mathrm{X}^\top \mathrm{e} = 0$.
- Under (A2), $\mathrm{X}^\top \mathrm{X} \succ 0$ is invertible: $\hat{\boldsymbol{\beta}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top \mathrm{y}$.
- Hessian $2\mathrm{X}^\top \mathrm{X} \succ 0 \Rightarrow$ unique global minimum (convex problem).

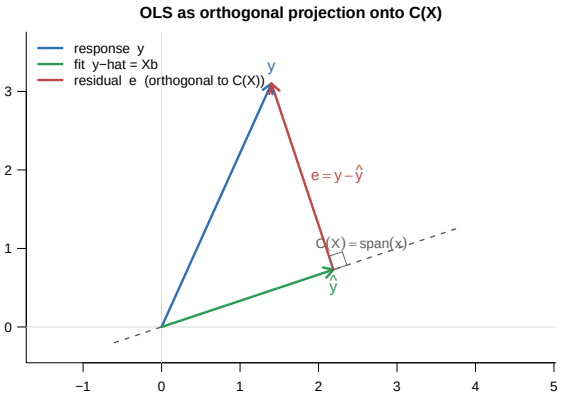
OLS is an orthogonal projection

$$\hat{y} = X\hat{\beta} = \underbrace{X(X^\top X)^{-1}X^\top}_{H} y = Hy.$$

The hat matrix H projects y onto $\mathcal{C}(X)$. The right angle in the figure *is* the normal equation $X^\top e = 0$.

Properties to know:

- $H^\top = H$ (symmetric)
- $HH = H$ (idempotent)
- $\text{tr}(H) = p$
- $M = I - H$ projects onto $\mathcal{C}(X)^\perp$, $\text{tr}(M) = n - p$



Leverage & a numerical sanity check

- Diagonal entries h_{ii} of H are leverages: $h_{ii} \in [0, 1]$, $\sum_i h_{ii} = p$.
- Leverage measures how far unit i's regressor row sits from the data centre.
- It drives the **outlier / influence** diagnostics of Lecture 3.

`swiss` model, verified in R

symmetric	TRUE
idempotent $HH = H$	TRUE
$\mathrm{tr}(H) = p$	$4 = 4$
$\max \lvert X^\top e \rvert$	$\approx 5 \times 10^{-11}$
leverage range	[0.033, 0.454]

ANOVA is Pythagoras in $\mathbb{R}^n$

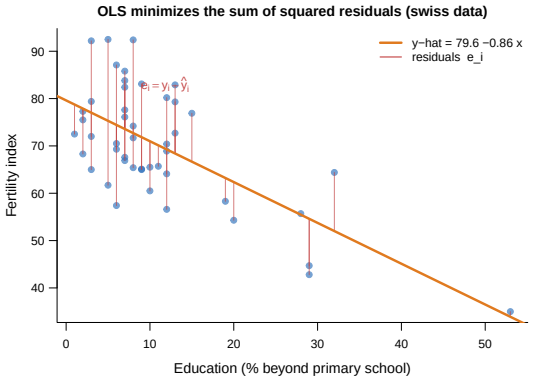
With an intercept, $1 \in \mathcal{C}(\mathrm{X})$, so $(\hat{\mathrm{y}} - \bar{y}1) \perp (\mathrm{y} - \hat{\mathrm{y}})$:

$$\underbrace{\|\mathrm{y} - \bar{y}1\|^2}_{\mathrm{SST}} = \underbrace{\|\hat{\mathrm{y}} - \bar{y}1\|^2}_{\mathrm{SSR}} + \underbrace{\|\mathrm{e}\|^2}_{\mathrm{SSE}}.$$

$$R^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = \cos^2\theta,$$

the squared cosine of the angle between centered y and centered $\hat{\mathrm{y}}$.

`swiss`: SST$= 7178$, SSR$= 4610.1$, SSE$= 2567.9$;
$R^2 = 0.6423$ (matches `summary(lm)`).



Frisch–Waugh–Lovell: what "controlling for" means

Partition $X = [X_1 \;\; X_2]$. The OLS estimate of β_1 equals:

1. residualize X_1 on X_2: $\tilde{X}_1 = M_2 X_1$;
2. residualize y on X_2: $\tilde{y} = M_2 y$;
3. regress $\tilde{y}$ on $\tilde{X}_1$.

Interpretation

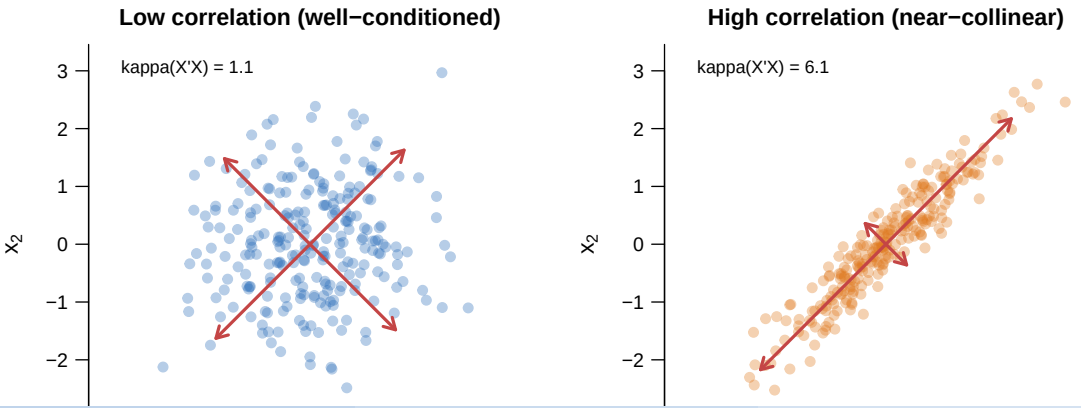
A multiple-regression coefficient relates y to a predictor **after partialling out the other predictors from both**. This is why omitting a relevant control changes a coefficient, and it underlies added-variable plots.

Verified in R: `Education` coef in full `swiss` model $= -1.0721$ equals the FWL slope to 8 decimals.

Eigenstructure $\Rightarrow$ multicollinearity

$\mathrm{X}^\top\mathrm{X}$ is symmetric PD with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p > 0$.

- $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathrm{X}^\top\mathrm{X})^{-1}$ is large in small-eigenvalue directions.
- Near-dependent columns $\Rightarrow$ a tiny λ_{min} $\Rightarrow$ unstable $\hat{\boldsymbol{\beta}}$.
- Condition number $\kappa = \sqrt{\lambda_{\text{max}}/\lambda_{\text{min}}}$; rule of thumb $\kappa \gtrsim 30$ is trouble.



From geometry to diagnostics

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}, \qquad R_j^2 \text{ from regressing predictor } j \text{ on the rest.}$$

`longley` (collinear macro data)

VIF: GNP ≈ 1034, Year ≈ 733, Population ≈ 226; $\kappa \approx 81$.

`swiss` (mild)

all VIF < 4; $\kappa \approx 2.4$.

Helpers: `vif_manual()`, `condition_number()` (in `R/00_helpers.R`). Detecting & remedying collinearity recurs in Lectures 5–6.

Compute with decompositions, not inverses

- **Direct inverse** $(X^\top X)^{-1} X^\top y$: transparent, but forming $X^\top X$ squares κ.
- **QR**: $X = QR$, solve $R\hat{\beta} = Q^\top y$ (R's `lm` default).
- **SVD**: $X = UDV^\top$, most robust; singular values expose conditioning; gives the pseudo-inverse when (A2) fails.

Verified

Direct, QR, SVD, and `lm` agree on `swiss` to $\sim 10^{-13}$.

Habit: derive with the closed form, compute with a tested solver.

Summary & what's next

One picture to remember

Mean response lives in $\mathcal{C}(\mathrm{X})$; OLS projects y orthogonally onto it (H); residuals live in $\mathcal{C}(\mathrm{X})^{\perp}$; ANOVA is Pythagoras; "controlling for" is FWL; stability is governed by the eigenvalues of $\mathrm{X}^{\top}\mathrm{X}$.

- **Lecture 2:** linear vs. nonlinear models; matrix representation of model families.
- **Lecture 3:** add the statistical assumptions $\Rightarrow$ Gauss–Markov, sampling distributions, tests, CIs (incl. bootstrap), diagnostics.
- **Lab 1:** do the matrix mechanics by hand and verify against `lm`.